Supplementary Table 1. Artificial Intelligence Models for Arrhythmia Detection

Study	Target disease (Task)	Dataset	Method	Results
Huang, et al. 2021[14]	Arrhythmia (Heartbeat classification)	MITADB	CNN and LSTM	Accuracy, 98.93%; Sensitivity, 96.46%; Specificity, 99.33%
Shi, et al. 2021[15]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	CNN	Accuracy, 91.7%; Specificity, 93.2%; Sensitivity, 90.4%; F1 score: 0.883
Yu, et al. 2021[16]	Arrhythmia (Detection of premature ventricular contraction)	MITADB	K-Nearest Neighbor	Accuracy, 99.7%; Sensitivity, 97.45%; Specificity, 99.87%
Jo, et al. 2021[17]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB and open database for external validation (PTB-XL, Chapman ECG DB and PhysioNet/ CinC Challenge 2017 DB)	CNN	Internal validation: (AUC, 0.994; Accuracy, 99.4%; Sensitivity, 98.2%; Specificity, 99.5%), External validation: (AUC, 0.996; Accuracy, 99.2%; Sensitivity, 99.2%; Specificity, 99.2%)
Lu, et al. 2021[18]	Arrhythmia (Identification of five different types of heartbeats)	MITADB	CNN	Accuracy, 99.31%; Sensitivity, 99.45%
Lee, et al. 2021[19]	Arrhythmia (Identification of four different types of heartbeats)	PhysioNet/ CinC Challenge 2017 DB	CNN	F1 score of normal rhythm detection, 0.8973; F1 score of AF detection, 0.8106; F1 score of other rhythms detection, 0.7445; F1 score of noise detection, 0.6222
Zhang, et al. 2021[20]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB and MITAFDB	CNN and LSTM	Accuracy, 95.44%; Sensitivity, 97.73%; Specificity, 95.19%
Yang, et al. 2021[21]	Arrhythmia (Detection of premature beats)	MITADB and CPSCDB	CNN	PVC detection: (Sensitivity, 95.8%; F1-score: 92.6%), SPB detection: (Sensitivity, 84.7%; F1-score: 72.2%)
Nguyen, et al. 2021[22]	Arrhythmia (Recognition of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	CNN and SVM	Sensitivity, 75.88%; F1 score: 84.19%; Precision, 81.65%
Liu, et al. 2021[23]	Arrhythmia (Prediction for tachycardia onset)	MIMIC-III and proprietary DB	Bidirectional LSTM	AUC, 0.856; Accuracy, 79.6%; Sensitivity, 77.6%; Specificity, 81.6%; F1 score: 79.1%
Tan, et al. 2021[24]	Arrhythmia (Detection of premature beats)	MITADB and CPSCDB	CNN	PVC detection: (Sensitivity, 95.8%; F1-score: 92.6%), SPB detection: (Sensitivity, 84.7%; F1-score: 72.2%)
Gahungu, et al. 2021[25]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB(1839 1-lead ECG)	CNN	Sensitivity, 100%; Specificity, 75.99%
Herraiz, et al. 2020[26]	Arrhythmia (Detection of Atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB, THDB+proprietary DB	CNN	Public dataset: (Accuracy 88.95%; Sensitivity 86.91%; Specificity 91.00%), Proprietary dataset: (Accuracy 94.79%; Sensitivity 97.17%; Specificity 92.42%)
Jeon, et al. 2020[27]	Arrhythmia (ECG beat classification)	MITADB+proprietary DB	RNN	Accuracy, 99.80%
Chen, et al. 2020[28]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (30,136 PPG+1-lead ECG)	CNN	Accuracy, 90.52% (95% CI, 87.24%–93.04%); Sensitivity, 80.00% (95% CI, 72.52%–85.90%); Specificity, 96.81% (95% CI, 93.58%–98.51%)
Gao, et al. 2020[29]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	CNN	Accuracy, 82.32%±0.87%
Marinucci, et al. 2020[30]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (8028 1-lead ECG)	ANN	AUC, 0.908 (95% CI, 0.881–0.935); Sensitivity, 81.2%; Specificity, 81.2%
Aschbacher, et al. 2020[31]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (51 patients PPG)	RNN	AUC, 0.983; Sensitivity, 98.5%; Specificity, 88.0%
Zhang, et al. 2020[32]	Arrhythmia (Screening of atrial fibrillation)	MITAFDB	CNN	Accuracy, 98.43%; Sensitivity, 98.10%; Specificity, 98.76%
Liang, et al. 2020[33]	Arrhythmia (Classification of arrhythmia)	MITADB	Genetic algorithm and SVM	Sensitivity, 90%; Specificity, 93%; F1 score: 90%
Ghosh, et al. 2020[34]	Arrhythmia (Detection of atrial fibrillation)	MITAFDB and MITADB	Hierarchical extreme learning machine classifier	Accuracy, 99.40%; Sensitivity, 98.77%; Specificity, 100%
Li, et al. 2020[35]	Arrhythmia (Identification of five different types of heartbeats)	MITADB	RNN	Accuracy, 99.06%
Yan, et al. 2020[36]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (320 facial PPG)	CNN	Accuracy, 95.9%; Sensitivity, 93.8%; Specificity, 98.1%
Vijayarangan, et al. 2020[37]	Arrhythmia (Classification of 8 types of heartbeats)	MITADB,LTAFDB and MITLTDB	CNN and LSTM	Accuracy, 97%; Sensitivity, 97%; F1 score: 0.97
Cheng, et al. 2020[38]	Arrhythmia (Multi-label ECG classification)	MITADB	CNN	Accuracy, 93.19–98.16%
Torres-Soto, et al. 2020[39]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB and 2015 IEEE Signal Processing Cup Dataset	CNN	Sensitivity, 98%; Specificity, 99%; F1 score: 0.96
Lai, et al. 2020[40]	Arrhythmia (Screening of atrial fibrillation)	Proprietary DB (1-lead ECG of 55 persons)	CNN	Accuracy, 93.1%; Sensitivity, 93.1%; Specificity, 93.4%
Jacobsen et al 2020[41]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (ECG of 102 persons)	RNN	AUC, 0.97; Sensitivity, 95.2%; Specificity, 92.5%
Lennox and Mahmud 2020[42]	Arrhythmia (ECG classification)	MITADB	ANN	Accuracy, 96.044%
Abdelazez, et al. 2020[43]	Arrhythmia (Detection of atrial fibrillation)	LTAFDB and MITAFDB	CNN	AUC, 0.87–0.75
Kwon, et al. 2019[44]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (13,038 PPG)	CNN	Accuracy, 96.9%; Sensitivity, 99.0%; Specificity, 94.3%
Wu, et al. 2019[45]	Arrhythmia (Detection of atrial fibrillation)	MITADB, MITMVADB, MITAFDB, LTAFDB, MITNSRDB and MITNST	CNN	AUC, 0.9983; Accuracy, 97.56%; Sensitivity, 97.56%; Specificity, 99.19%
Kwon, et al. 2019[46]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (30,103 PPG)	CNN	AUC, 0.998 (95% CI, 0.995–1.000); Accuracy, 97.6%; Sensitivity, 99.3%; Specificity, 95.9%
Hannun, et al. 2019[47]	Arrhythmia (Classification of arrhythmia)	Proprietary DB (91232 1-lead ECG)	CNN	Overall: (AUC, 0.97; F1 score: 0.837), AF detection: (Sensitivity, 90.8%; Specificity, 90.6%)
Hong, et al. 2019[48]	Arrhythmia (Classification of arrhythmia)	PhysioNet/ CinC Challenge 2017 DB	CNN	F1 score of normal rhythm detection, 0.9117; F1 score of AF detection, 0.8128; F1 score of other rhythms detection, 0.7505; F1 score of noise detection, 0.567
Lai, et al. 2019[49]	Arrhythmia (Detection of atrial fibrillation)	MITAFDB	CNN	Accuracy, 97.3%; Sensitivity, 97.4%; Specificity, 97.2%
Kwon, et al. 2019[50]	Arrhythmia (Detection of arrhythmia)	Proprietary DB (PPG of 75 patients)	CNN	AUC, 0.998; Accuracy, 97.58%; Sensitivity, 99.32%; Specificity, 95.58%
Amirshahi and Hashemi 2019[51]	Arrhythmia (ECG beat classification)	MITADB	Spiking neural network	Accuracy, 97.9%; Sensitivity, 80.2%; Specificity, 99.8%
Wang, et al. 2019[52]	Arrhythmia (Detection of cardiac arrhythmia)	MITADB	CNN	VEB detection: (Accuracy, 99.1; Sensitivity, 92.8; Specificity, 99.6), SVEB detection: (Accuracy, 99.5; Sensitivity, 88.0; Specificity, 99.8)
Wasserlauf, et al. 2019[53]	Arrhythmia (Detection and quantification of atrial fibrillation)	proprietary DB (7500 1-lead ECG+1-lead ECG of 26 persons)	CNN	Sensitivity, 97.7%; Specificity, 98.9%
Sadr, et al. 2018[54]	Arrhythmia (ECG beat classification)	PhysioNet/ CinC Challenge 2017 DB	Neural network with quadratic function	F1 score of normal rhythm detection, 0.90; F1 score of AF detection, 0.75; F1 score of other rhythms detection, 0.68; F1 score of noise detection, 0.32
Mathews, et al. 2018[55]	Arrhythmia (Classify the two critically abnormal eartbeats)	MITADB	Restricted boltzmann machine and deep belief network	Accuracy of VEB detection, 93.63%; Accuracy of SVEB detection, 95.57%
Kamaleswaran, et al. 2018[56]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	CNN	Accuracy, 84.24% (95% CI, 83.32%–84.71%); F1-score: 0.83
Teijeiro, et al. 2018[57]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	Tree gradient boosting model and RNN	F1 score, 0.85
Hernandez, et al. 2018[58]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	ANN	F1 score, 0.64
Rubin, et al. 2018[59]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB and proprietary DB	CNN	F1 score, 0.82
Khamis, et al. 2018[60]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	ANN	F1 score, 0.80
Xiong, et al. 2018[61]	Arrhythmia (Detection of cardiac arrhythmia)	PhysioNet/ CinC Challenge 2017 DB	RNN	F1 score, 0.82
Fan, et al. 2018[62]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	CNN	Accuracy 98.13%; Sensitivity, 93.77%; Specificity, 98.77%
Plesinger, et al. 2018[63]	Arrhythmia (Classification of holter ECG)	PhysioNet/ CinC Challenge 2017 DB	CNN and bagged tree ensemble	F1 score of normal rhythm detection, 0.92; F1 score of AF detection, 0.82; F1 score of other rhythms detection: 0.74
Tison, et al. 2018[64]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (12-lead ECG of 11417 persons)	ANN	Cardioversion cohort: (AUC, 0.97; Sensitivity, 98.0%; Specificity, 90.2%), Ambulatory subcohort: (AUC, 0.72; Sensitivity, 67.7%; Specificity, 67.6%)
Zhou, et al. 2018[65]	Arrhythmia (Detection of premature ventricular contraction)	MITADB	RNN	Accuracy, 96%–99%; Sensitivity, 99%–100%; Specificity, 94%–96%
Yan, et al. 2018[66]	Arrhythmia (Detection of atrial fibrillation)	Proprietary DB (PPG of 51 persons)	ANN	AUC, 0.991; Sensitivity, 93.3%; Specificity, 94.4%
Clifford, et al. 2017[67]	Arrhythmia (Detection of atrial fibrillation)	PhysioNet/ CinC Challenge 2017 DB	Voting method	F1 score: 0.868
Kiranyaz, et al. 2016[68]	Arrhythmia (Real-time classification of ECGs)	MITADB	CNN	VEB detection: (Accuracy, 99%; Sensitivity, 93.9%; Specificity, 98.9%), SVEB detection: (Accuracy, 97.6%; Sensitivity, 60.3%; Specificity, 99.2%)
Liu, et al. 2015[69]	Arrhythmia (Detection of premature ventricular contraction)	MITADB	ANN	Accuracy, 98.90%; Sensitivity, 90.26%; Specificity, 92.32%
Solosenko, et al. 2015[70]	Arrhythmia (Detection of premature ventricular contraction)	MIMIC II	ANN	Normal detection: (Accuracy, 99.7%; Sensitivity, 99.8%; Specificity, 93.3%), PVC 1 detection: (Accuracy, 99.8%; Sensitivity, 93.2%; Specificity, 99.9%), PVC 2 detection: (Accuracy, 99.8%; Sensitivity, 92.4%; Specificity, 99.9%)
Liu, et al. 2013[71]	Arrhythmia (Classification of arrhythmia)	MITADB	SVM and ANN	Accuracy, 96.4%
Oresko, et al. 2010[72]	Arrhythmia (Detection of arrhythmia)	MITADB	ANN	Accuracy, 93.32%
Jin, et al. 2009[73]	Arrhythmia (Prediction of rhythm abnormalities)	Proprietary DB and MITADB	ANN	Accuracy of normal rhythm detection, 99%; Accuracy of Right bundle branch block beat detection, 98%; Accuracy of PVC detection, 92.6%; Accuracy of Paced beat detection, 96%; Accuracy of Fusion of paced and normal beat detection, 81%
Rodriguez, et al. 2005[74]	Arrhythmia (Real-time classification of ECGs)	MITADB,LTSTDB, EST-TDB, MITNST and ANSI/AAMI EC13	Decision tree	Accuracy, 100%
Hickey, et al. 2004[75]	Arrhythmia (Assessment of paroxysmal atrial fibrillation)	AFPDB, NSRDB, UCDODB and MITAFDB	Linear discriminant classifier	Accuracy, 80.5%; Sensitivity, 43.0%; Specificity, 99.3%

ECG, electrocardiography; PPG, photoplethysmography; ANN, Artificial neural network; AF, Atrial fibrillation; AUC, Area under the curve; CNN, Convolutional neural network; LSTM, Long short term memory; PVC, Premature ventricular contraction; SPB, Supraventricular premature beat; SVEB, Supra ventricular ectopic beat; SVM, Support vector machine; RNN, Recurrent neural network.
Refer to Table 2 for artificial intelligence models for cardiovascular-related diseases other than arrhythmia. Refer to Supplementary Table 2 (only online) for dataset abbreviations and description.